

MATHEMATICS

TARGET : JEE- Advanced

CAPS-25
CONIC SECTIONS**SCQ (Single Correct Type) :**

- If α, β are eccentric angles of the extremities of a focal chord of an ellipse, then eccentricity of the ellipse is
 (A) $\frac{\cos \alpha + \cos \beta}{\cos(\alpha + \beta)}$ (B) $\frac{\sin \alpha - \sin \beta}{\sin(\alpha - \beta)}$ (C) $\sec \alpha + \sec \beta$ (D) $\frac{\sin \alpha + \sin \beta}{\sin(\alpha + \beta)}$
- A series of concentric ellipses $E_1, E_2, \dots, E_n$ are drawn such that E_n touches the extremities of the major axis of E_{n-1} and the foci of E_n coincide with the extremities of minor axis of E_{n-1} . If the eccentricity of the ellipses is independent of n , then the value of the eccentricity, is
 (A) $\frac{\sqrt{5}}{3}$ (B) $\frac{\sqrt{5}-1}{2}$ (C) $\frac{\sqrt{5}+1}{2}$ (D) $\frac{1}{\sqrt{5}}$
- The transverse axis of a hyperbola is of length $2a$ and a vertex divides the segment of the axis between the centre and the corresponding focus in the ratio $2 : 1$, the equation of the hyperbola is :
 (A) $4x^2 - 5y^2 = 4a^2$ (B) $4x^2 - 5y^2 = 5a^2$ (C) $5x^2 - 4y^2 = 4a^2$ (D) $5x^2 - 4y^2 = 5a^2$
- If AB is a double ordinate of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ such that $\triangle OAB$ (O is the origin) is an equilateral triangle, then the eccentricity 'e' of the hyperbola satisfies
 (A) $e > \sqrt{3}$ (B) $1 < e < 2\frac{2}{\sqrt{3}}$ (C) $e = \frac{2}{\sqrt{3}}$ (D) $e > \frac{2}{\sqrt{3}}$
- An ellipse whose major axis is parallel to X-axis such that the segments of the focal chords are 1 and 3 units. The lines $ax + by + c = 0$ are the chords of the ellipse such that a, b, c are in A.P. and bisected by the point at which they intersect. The equation of its auxiliary circle is $x^2 + y^2 + 2\alpha x + 2\beta y - 2\alpha - 1 = 0$ then _____.
 Equation of the auxiliary circle is
 (A) $x^2 + y^2 - 2x + 4y + 1 = 0$ (B) $x^2 + y^2 + 2x + 2y - 3 = 0$
 (C) $x^2 + y^2 + 2x + 4y + 1 = 0$ (D) $x^2 + y^2 - 4x + 2y - 3 = 0$

MCQ (One or more than one correct) :

- A point moves such that the sum of the squares of its distances from the two sides of length 'a' of a rectangle is twice the sum of the squares of its distances from the other two sides of length 'b'. The locus of the point can be :
 (A) a circle (B) an ellipse (C) a hyperbola (D) a pair of lines



7. Which of the following equations in parametric form can represent a hyperbolic profile, where 't' is a parameter.
- (A) $x = \frac{a}{2} \left(t + \frac{1}{t} \right)$ & $y = \frac{b}{2} \left(t - \frac{1}{t} \right)$ (B) $\frac{tx}{a} - \frac{y}{b} + t = 0$ & $\frac{x}{a} + \frac{ty}{b} - 1 = 0$
- (C) $x = e^t + e^{-t}$ & $y = e^t - e^{-t}$ (D) $x^2 - 6 = 2 \cos t$ & $y^2 + 2 = 4 \cos^2 \frac{t}{2}$
8. Let $C_1 : 9x^2 - 16y^2 - 18x + 32y - 23 = 0$ and $C_2 : 25x^2 + 9y^2 - 50x - 18y + 33 = 0$ are two conics then
- (A) eccentricity of C_1 is $\frac{5}{4}$. (B) eccentricity of C_2 is $\frac{5}{3}$.
- (C) area of the quadrilateral with vertices at the foci of the conics is $\frac{8}{9}$.
- (D) latus rectum of C_1 is greater than latus rectum of C_2 .
9. A hyperbola centred at C has one focus at P(6, 8). If its directrices are $3x + 4y + 10 = 0$ and $3x + 4y - 10 = 0$, then ____.
- (A) CP = 10 (B) eccentricity = $\sqrt{5}$
- (C) CP = 8 (D) eccentricity = $\frac{\sqrt{5}}{2}$
10. Two ellipses $\frac{x^2}{\cos^2 \alpha} + \frac{y^2}{\sin^2 \alpha} = 1$ and $\frac{x^2}{\sin^2 \alpha} + \frac{y^2}{\cos^2 \alpha} = 1$ ($0 < \alpha < \frac{\pi}{4}$) intersect at four points P, Q, R, S then which of the following statement(s) is /are true?
- (A) PQRS is a square with length of the side $\sin 2\alpha$
- (B) PQRS lie on a circle whose centre is origin and with radius $\frac{\sin 2\alpha}{\sqrt{2}}$
- (C) eccentricity of the two given ellipses are same
- (D) there are two points on $\frac{x^2}{\sin^2 \alpha} + \frac{y^2}{\cos^2 \alpha} = 1$ whose reflection in $y = x$ lie on the same ellipse

Comprehension Type Question:**Comprehension # 1****Paragraph for question nos. 11 to 13**

The graph of the conic $x^2 - (y - 1)^2 = 1$ has one tangent line with positive slope that passes through the origin. the point of tangency being (a, b). Then

11. The value of $\sin^{-1} \left(\frac{a}{b} \right)$ is
- (A) $\frac{5\pi}{12}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$
12. Length of the latus rectum of the conic is
- (A) 1 (B) $\sqrt{2}$ (C) 2 (D) none

13. Eccentricity of the conic is

(A) $\frac{4}{3}$

(B) $\sqrt{3}$

(C) 2

(D) none

Numerical based Questions :

14. Common tangents are drawn to the parabola $y^2 = 4x$ & the ellipse $3x^2 + 8y^2 = 48$ touching the parabola at A & B and the ellipse at C & D, then the area of the quadrilateral ABCD is $\lambda\sqrt{2}$ the λ is equal to

15. If two points P & Q on the hyperbola $x^2/a^2 - y^2/b^2 = 1$ whose centre is C be such that CP is perpendicular to CQ & $a < b$, then $\frac{1}{CP^2} + \frac{1}{CQ^2} = \lambda \left(\frac{1}{a^2} - \frac{1}{b^2} \right)$ where λ is :

16. If the distance between the centres of the hyperbolas :

$x^2 - 16xy - 11y^2 - 12x + 6y + 21 = 0$ (i)

$9x^2 - 16y^2 - 18x - 32y - 151 = 0$ (ii)

is d then $125d^2 = \dots\dots\dots$

17. If $(a^2, a - 2)$ be a point interior to the region of the parabola $y^2 = 2x$ bounded by the chord joining the points (2, 2) and (8, -4), then the number of all possible integral values of a is :

18. If a tangent of slope 2 of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is normal to the circle $x^2 + y^2 + 4x + 1 = 0$, then maximum value of ab is _____.

19. If the length of the latus rectum of a standard hyperbola of eccentricity 2 is equal to the limit of the series $\frac{3}{1^2} + \frac{5}{1^2 + 2^2} + \frac{7}{1^2 + 2^2 + 3^2} + \dots$ and r is the radius of the director circle of its conjugate hyperbola, then $r^2 = \dots\dots\dots$.

20. Let $C_1 : x^2 - y^2 = 5$ and $C_2 : x^2 + y^2 - 8y + 3 = 0$ be the equations of a hyperbola and a circle respectively. The curves C_1 and C_2 touch each other at $(\pm 3, 2)$. $P(0, 4 - \sqrt{13})$ is a point on the curve C_2 . Let a line through P meet C_1 at m number of points and C_2 at n number of points. If $(m + n) = 3$, then the number of such straight lines is _____.

21. Coordinates of the vertices B and C are (2, 0) and (8, 0) respectively. The vertex A is varying in such a way that $4 \tan \frac{B}{2} \tan \frac{C}{2} = 1$. If the locus of A is an ellipse then the length of its semi major axis is _____.

Matrix Match Type :

22. Match the following:

$$\text{For the ellipse } (3x - 6)^2 + (3y - 9)^2 = \frac{4}{169}(5x + 12y + 6)^2$$

Column-I	Column-II
(a) Length major axis	(p) $\frac{24}{5}$
(b) Length minor axis	(q) $\frac{16}{5}$
(c) Length of Latus Rectum	(r) $\frac{16}{3}$
(d) Distance between directrices	(s) $\frac{72}{5}$
	(t) $\frac{48}{5}$

(A) $a \rightarrow r$; $b \rightarrow q$; $c \rightarrow r$; $d \rightarrow s$ (B) $a \rightarrow t$; $b \rightarrow q$; $c \rightarrow r$; $d \rightarrow s$ (C) $a \rightarrow t$; $b \rightarrow s$; $c \rightarrow r$; $d \rightarrow s$ (D) $a \rightarrow t$; $b \rightarrow q$; $c \rightarrow q$; $d \rightarrow s$ **Subjective Type Questions :**

23. If S and H be the foci of an ellipse and any point A be taken on the curve and the chords ASB, BHC, CSD and DHE be drawn and eccentric angles of A, B, C, D, E, be $\theta_1, \theta_2, \theta_3, \theta_4, \dots$. Prove that $\tan \frac{\theta_1}{2} \tan \frac{\theta_2}{2} = \cot \frac{\theta_2}{2} \cot \frac{\theta_3}{2} = \tan \frac{\theta_3}{2} \tan \frac{\theta_4}{2} = \dots$
24. A variable point P on an ellipse of eccentricity e is joined to its foci S, S'. Prove that the locus of the incentre of $\triangle PSS'$ is an ellipse of eccentricity $\sqrt{\frac{2e}{1+e}}$.
25. Let P and Q be two points on the ellipse $x^2 + 4y^2 = 16$, whose eccentric angles are $\frac{\pi}{4}$ & $\frac{3\pi}{4}$ respectively. The tangent at P and the normal at Q cut each other at R and the normal at Q cuts the ellipse again at M. Find the area of the triangle PRM.
26. A rectangular hyperbola, with centre C, is intersected by a circle of radius r in four points p, Q, R and S. Prove that $CP^2 + CQ^2 + CR^2 + CS^2 = 4r^2$

ANSWER KEY OF CAPS-25

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|---|---|---|----------|------------|
| 1. (D) | 2. (B) | 3. (D) | 4. (D) | 5. (A) |
| 6. (CD) | 7. (ACD) | 8. (ACD) | 9. (AB) | 10. (ABCD) |
| 11. (D) | 12. (C) | 13. (D) | 14. (55) | 15. (1) |
| 16. (25) | 17. (1) | 18. (4) | 19. (2) | 20. (7) |
| 21. (5) | 22. (B) | 23. $\tan \frac{\theta_1}{2} \tan \frac{\theta_2}{2} = \cot \frac{\theta_2}{2} \cot \frac{\theta_3}{2} = \tan \frac{\theta_3}{2} \tan \frac{\theta_4}{2} = \dots \text{etc.}$ | | |
| 24. $\sqrt{\frac{2e}{1+e}}$ | 25. $\Delta = \frac{2}{17} \times 2(17 + 23) = \frac{4}{17} \times 40 = \frac{160}{17}$ sq. units | | | |
| 26. $= 4r^2 + 4h^2 + 4k^2 - 2h \cdot 2h - 2k \cdot 2k = 4r^2$ | | | | |